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SHORT COMMUNICATION

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Inter-habitat community variation of butterflies in an Afrotropical region

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Abstract

The impact of habitat disturbance on butterfly species assemblage in the Afrotropical region was studied by investigating inter-habitat community variation of butterflies. Distinct butterfly species assemblages were found across surveyed habitat types, with 1554 adult butterflies of 126 species from five families recorded. The Primary forests had the highest species diversity and a higher dissimilarity between the Secondary and Plantation Forests. Of the 126 species recorded, 80 species were unique to one habitat. We provide new insights into butterfly ecology in the region, which can inform the development of conservation strategies to protect these ecosystems.

Abstract

L'impact des perturbations de l'habitat sur l'assemblage des espèces de papillons dans la région afrotropicale a été étudié en examinant les variations des communautés de papillons entre les habitats. Des assemblages différents d'espèces de papillons ont été observés dans tous les types d'habitats étudiés, avec 1 554 papillons adultes de 126 espèces appartenant à cinq familles. Les forêts vierges présentaient la plus grande diversité d'espèces et une plus grande dissimilitude entre les forêts secondaires et les forêts de plantation. 80 sur les 126 espèces enregistrées étaient uniques à un habitat. Nous fournissons de nouvelles informations sur l'écologie des papillons dans la région, afin de contribuer à l'élaboration de stratégies de conservation visant à protéger ces écosystèmes.

1 | INTRODUCTION

Habitat disturbance poses a significant threat to butterfly diversity and conservation, altering forest ecosystems' structure, composition, and function, thereby affecting the availability and quality of resources and microclimates for butterflies (Habel et al., 2018). Fragmentation resulting from disturbance further diminishes the connectivity and size of suitable butterfly habitats (Attiwilli et al., 2022). Due to their sensitivity to environmental changes, butterflies are important bioindicators for ecosystem health and biodiversity conservation. Despite the remarkable diversity of butterfly communities in the Afrotropical region, more knowledge is needed regarding community variations across different habitats, particularly in response to ecological disturbances.

Understanding the factors that shape butterfly community composition and structure is crucial for effective species and ecosystem conservation. Various factors, including habitat quality, geographic location, and topography, contribute to the characteristics and compositions of butterflies in ecological communities (Kocher & Williams, 2000). Habitat disturbance, leading to reduced habitat quality, is widely recognised as a significant driver of butterfly species richness and abundance decline in diverse ecological communities (Brown & Hutchings, 1997; Hamer et al., 1997; Spitzer et al., 1993, 1997).

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Human activities such as agriculture, urbanisation, and logging significantly alter or degrade natural habitats, causing notable shifts in butterfly community composition and structure. Immediate impacts include the loss of native host plants and nectar sources, hindering butterfly populations. Disturbed habitats with non-native plant species may lack appropriate resources, further contributing to population declines (Hammond et al., 2019). Additionally, habitat disturbance alters environmental conditions that influence butterfly behaviour, physiology, and distribution, including temperature, humidity, and light availability (Satterfield et al., 2015). Fragmentation resulting from disturbance reduces butterfly habitats, leading to smaller and isolated populations with limited dispersal and gene flow, increasing the risk of genetic drift and inbreeding (Thomas et al., 2004). Consequently, genetic diversity is diminished, rendering butterfly populations more susceptible to extinction and underscoring the importance of habitat conservation for butterfly species.

Addressing the question of how ecological disturbances affect butterfly diversity in Afrotropical habitats is the central focus of this study. We aim to investigate the variation in butterfly species assemblages across habitats with different levels of ecological disturbances in the region. Our hypothesis posits that human-induced habitat disturbance will lead to an altered composition of butterfly species, highlighting human activities' significant influence on vegetation and butterfly communities. Thus, we predict that the disturbance caused by human activities primarily impacts the vegetation composition within each habitat, ultimately influencing the composition of butterfly species.

The findings from this study will provide valuable insights into butterfly ecology and contribute to guiding conservation efforts to protect biodiversity in Afrotropical ecosystems.

2 | MATERIALS AND METHODS

We surveyed seven sites in South-East Nigeria, representative of four mesic habitat types: Primary forests, Secondary forests, Plantation forests, and Farmlands. The habitats were classified based on vegetation composition, land use, and protection status (Table 1). Primary forests were remnants of typical mature rainforests dominated by indigenous plant species, with some form of restriction from the most common human activities such as farming and logging. Secondary forests were sparsely wooded areas with regenerated forest patches; Plantation forests were managed for seedlings/timber production with native and exotic tree species, and Farmlands were cultivated areas for garden, food, and cash crops.

Butterfly species were recorded using the walk-and-count method along line transects (Pollard, 1977). Ten transects of 100 m were covered by three observers at each site and surveyed at least once during both the dry (November/December) and rainy (May/June) seasons, between 10:00 AM and 02:00 PM. All butterfly species in the habitats were photographed or captured using a sweep net and identified using Larsen (2005). Butterfly species encountered within 10 m along each transect were counted. We estimated biodiversity indices and performed cluster analysis using the vegan (Oksanen et al., 2022), cluster (Maechler et al., 2022) and "ComplexHeatmap" (Gu, 2022) packages in R version 4.2.0 (R Core Team, 2021).

3 | RESULTS

A total of 1554 adult butterfly individuals belonging to 126 species and five families were recorded across the surveyed habitat types. In all, six species belonged to the Family Papilionidae, 80 species to Family Nymphalidae, 15 species to Family Pieridae, 16 species to Family Lycaenidae and nine species to Family Hesperiidae. The five recorded butterfly families (Papilionidae, Nymphalidae, Pieridae, Lycaenidae, and Hesperiidae) were present in the Primary forests, Secondary Forests and Plantation Forests. There was no record of butterfly species belonging to the family Hesperiidae in the Farmlands (Table 2).

The Primary forests supported the highest butterfly species assemblage compared to the Secondary Forests, Plantation Forests and Farmlands based on the Shannon and Simpson Diversity indices (Table 3). The least species richness and abundance were observed

TABLE 1 Sites where butterflies were surveyed during the study.

Site label	Habitat type	Name of site/location	Coordinates
Site I	Primary forest	Eme River Forest Reserve, Bende, Abia State	05°38.5′ N; 007°31.4′ E
Site II	Farmland	Oguzoronweya Community Farms, Ebonyi State	06°27.3′ N; 008°07.6′ E
Site III	Plantation forest	Umuahia-Ibeku Forest Reserve, Umuahia, Abia State	05°31.9′ N; 007°29.0′ E
Site IV	Plantation forest	FRIN[a] Forest Outstation, Okwuta, Abia State	05°30.8′ N; 007°31.6′ E
Site V	Farmland	Omoba Community, Abia State	05°14.9′ N; 007°24.2′ E
Site VI	Secondary forest	ARCC[b] Ishiagu, Ebonyi State	05°52.0′ N; 007°33.9′ E
Site VII	Primary forest	MOUAU[c] Relict Forest, Umudike, Abia State	05°28.5′ N; 007°32.2′ E

[a]Forestry Research Institute of Nigeria.
[b]African Resource Conservation Centre.
[c]Michael Okpara University of Agriculture, Umudike.

TABLE 2 Records of butterfly species in various habitat types.

SN	Family	Species	Primary forests	Secondary forests	Plantation forests	Farmlands
1	Papilionidae	*Graphium angolanus baronis*			3	3
2	Papilionidae	*Graphium leonidas leonidas*				2
3	Papilionidae	*Graphium liponesco*			1	2
4	Papilionidae	*Papilio dardanus dardanus*			1	4
5	Papilionidae	*Papilio demodocus demodocus*	4	3	24	12
6	Papilionidae	*Papilio sosia*			8	3
7	Nymphalidae	*Acraea abdera eginopsis*	15		12	
8	Nymphalidae	*Acraea acerata*	28	5		
9	Nymphalidae	*Acraea alciope*	16			
10	Nymphalidae	*Acraea bonasia*		2		
11	Nymphalidae	*Acraea camaena*	33	6		21
12	Nymphalidae	*Acraea circeis*	3			
13	Nymphalidae	*Acraea egina egina*	4			1
14	Nymphalidae	*Acraea encedon encedon*	22			3
15	Nymphalidae	*Acraea esebria*		1		
16	Nymphalidae	*Acraea jodutta*	1			
17	Nymphalidae	*Acraea lycoa lycoa*		2		
18	Nymphalidae	*Acraea parrhasia*		3		
19	Nymphalidae	*Acraea pharsalus*				1
20	Nymphalidae	*Acraea pseudegina*		2	11	2
21	Nymphalidae	*Acraea serena*	40	8	28	8
22	Nymphalidae	*Acraea tellus tellus*			9	
23	Nymphalidae	*Acraea umbra*			4	
24	Nymphalidae	*Amauris hecate Hecate*		3		2
25	Nymphalidae	*Aterica galene galena*	4			4
26	Nymphalidae	*Bebearia cocalia continentalis*	7			
27	Nymphalidae	*Bebearia maledicta*	4			
28	Nymphalidae	*Bebearia sophus sophus*	2			
29	Nymphalidae	*Bebearia tentyris*	6			
30	Nymphalidae	*Bicyclus abnormis*				4
31	Nymphalidae	*Bicyclus dorothea dorothea*	10			
32	Nymphalidae	*Bicyclus safitza safitza*	2			
33	Nymphalidae	*Bicyclus sandace*				1
34	Nymphalidae	*Bicyclus taenias*				3
35	Nymphalidae	*Bicyclus xeneas occidentalis*				2
36	Nymphalidae	*Byblia anvatara crameri*		3		
37	Nymphalidae	*Catacroptera cloanthe ligata*			1	
38	Nymphalidae	*Charaxes ameliae doumeti*				4
39	Nymphalidae	*Charaxes boueti*			3	4
40	Nymphalidae	*Charaxes fulvescens senegala*	6			
41	Nymphalidae	*Charaxes protoclea*			3	
42	Nymphalidae	*Charaxes varanes vologeses*	4			
43	Nymphalidae	*Cymothoe ogova*	3			
44	Nymphalidae	*Cymothoe preussi*	1			2
45	Nymphalidae	*Cyrestis camillus*	2	2		

(Continues)

TABLE 2 (Continued)

SN	Family	Species	Primary forests	Secondary forests	Plantation forests	Farmlands
46	Nymphalidae	*Danaus chrysippus chrysippus*	27	4	27	3
47	Nymphalidae	*Elymniopsis bammakoo*	1	1		
48	Nymphalidae	*Euphaedra acuta*				1
49	Nymphalidae	*Euphaedra harpalyce*			3	
50	Nymphalidae	*Euphaedra janetta*			6	
51	Nymphalidae	*Euphaedra medon medon*			15	2
52	Nymphalidae	*Euphaedra ruspina*			2	
53	Nymphalidae	*Euriphene ampedusa*	7			
54	Nymphalidae	*Euriphene coerulea*			1	
55	Nymphalidae	*Euriphene conjungens*	4	3		
56	Nymphalidae	*Euryphura plautilla*			5	
57	Nymphalidae	*Hallelesis asochis*	3			
58	Nymphalidae	*Hamanumida daedalus*				3
59	Nymphalidae	*Harma theobene*	3			
60	Nymphalidae	*Hypolimnas anthedon*	3		4	
61	Nymphalidae	*Hypolimnas misippus*			7	
62	Nymphalidae	*Hypolimnas salmacis*	5			
63	Nymphalidae	*Junonia chorimene*	3		5	
64	Nymphalidae	*Junonia hierta cebrene*			1	1
65	Nymphalidae	*Junonia oenone*	16	11	46	21
66	Nymphalidae	*Junonia sophia Sophia*	4			11
67	Nymphalidae	*Junonia stygia*	6		18	
68	Nymphalidae	*Junonia terea terea*	25	6	18	8
69	Nymphalidae	*Melanitis leda*		5		5
70	Nymphalidae	*Melanitis Libya*		3		6
71	Nymphalidae	*Neptidopsis ophione*	7			
72	Nymphalidae	*Neptis metella*	8		2	
73	Nymphalidae	*Neptis morosa*	14	1	12	3
74	Nymphalidae	*Neptis nemestes nemestes*			1	
75	Nymphalidae	*Neptis troundi*	2			
76	Nymphalidae	*Phalanta eurytis eurytis*	4			
77	Nymphalidae	*Phalanta phalanta aethiopica*				1
78	Nymphalidae	*Precis antilope*	17			
79	Nymphalidae	*Precis milonia milonia*	14			
80	Nymphalidae	*Precis octavia Octavia*		2		3
81	Nymphalidae	*Precis pelarga*	10			8
82	Nymphalidae	*Protogoniomorpha parhassus*	1			
83	Nymphalidae	*Pseudoneptis bugandensis ianthe*	3		2	
84	Nymphalidae	*Sevenia pechueli sangbae*	2		4	
85	Nymphalidae	*Tirumala petiverana*		10		
86	Nymphalidae	*Ypthima doleta*	5			
87	Pieridae	*Appias epaphia epaphia*	16	11	25	
88	Pieridae	*Appias sabina sabina*	29		21	21
89	Pieridae	*Belenois aurota*			4	
90	Pieridae	*Belenois calypso*	22		11	12

TABLE 2 (Continued)

SN	Family	Species	Primary forests	Secondary forests	Plantation forests	Farmlands
91	Pieridae	*Catopsilia florella*	25		49	
92	Pieridae	*Colotis antevippe*				10
93	Pieridae	*Colotis euippe*				2
94	Pieridae	*Eurema desjardinsii regularis*	40	8	50	
95	Pieridae	*Eurema hapale*	1			
96	Pieridae	*Eurema hecabe solifera*	7			2
97	Pieridae	*Leptosia alcesta*	47		29	24
98	Pieridae	*Mylothris chloris*	39	8	15	10
99	Pieridae	*Mylothris dimidiata*			2	
100	Pieridae	*Mylothris jaopura*	2			
101	Pieridae	*Mylothris sjoestedti sjoestedti*			1	
102	Hesperiidae	*Atalopedes campestris*		3		
103	Hesperiidae	*Coeliades forestan*			1	
104	Hesperiidae	*Eagris subalbida subalbida*			1	
105	Hesperiidae	*Pardaleodes edipus*			2	
106	Hesperiidae	*Pardaleodes incerta murcia*		2		
107	Hesperiidae	*Pelopidas mathias*		1		
108	Hesperiidae	*Sarangesa thecla*	2			
109	Hesperiidae	*Spialia ploetzi occidentalis*			2	
110	Hesperiidae	*Tagiades flesus*	2			
111	Lycaenidae	*Anthene larydas*	2			
112	Lycaenidae	*Anthene liodes*		2		
113	Lycaenidae	*Anthene radiate*			4	
114	Lycaenidae	*Aslauga bella*	3			
115	Lycaenidae	*Euchrysops barkeri*	4		3	1
116	Lycaenidae	*Geritola subargentea*	3			
117	Lycaenidae	*Hypolycaena anara*		2		
118	Lycaenidae	*Hypolycaena hatita*			2	
119	Lycaenidae	*Hypolycaena philippus*			1	
120	Lycaenidae	*Lepidochrysops dunni*		4		
121	Lycaenidae	*Lepidochrysops ringa*		2		
122	Lycaenidae	*Lepidochrysops synchrematiza*		3		
123	Lycaenidae	*Myrina silenus*	2			
124	Lycaenidae	*Oxylides faunus faunus*	2			
125	Lycaenidae	*Pseuderesia eleaza*			4	
126	Lycaenidae	*Thermoniphas micylus*	2	1		

in the Secondary Forests but not in species evenness. The butterfly species composition in Farmland was more similar to that of Plantation (66.35%) and Secondary forests (65.10%) than to that of Primary forests. Plantation forests had a slightly higher similarity with the Secondary forests (3.91%) compared to the Primary forests (1.18%). The Primary forests showed a 9.17% similarity in species composition with the Secondary forests (Figure 1). Of the 126 butterfly species recorded, 80 species were unique to one of the habitats. Thirty-two species were recorded only in the Primary Forests, 14 in Secondary Forests, 22 species in Plantation forests and 12 species in Farmlands. Furthermore, 31 species were recorded in at least two habitat types, eight species in three habitat types (*Acraea camaena*, *Acraea pseudegina*, *Appias epaphia epaphia*, *Appias sabina sabina*, *Belenois calypso*, *Eurema desjardinsii regularis*, *Leptosia alcesta*, *Euchrysops barkeri*) and seven species across the four surveyed habitat types (*Papilio demodocus demodocus*, *Acraea serena*, *Danaus chrysippus chrysippus*, *Junonia oenone*, *Junonia terea terea*, *Neptis morosa*, *Mylothris chloris*; Figure 1). Overall, butterfly species assemblages across the surveyed habitat types were above 66% distinct between pairs of habitats.

TABLE 3 Diversity indices and evenness across habitats.

Habitats	Shannon	Simpson	Richness	Abundance	Evenness
Primary forests	3.67	0.97	66	661	0.88
Secondary forests	3.29	0.95	34	133	0.93
Plantation forests	3.33	0.95	51	514	0.88
Farmlands	3.33	0.95	43	246	0.89

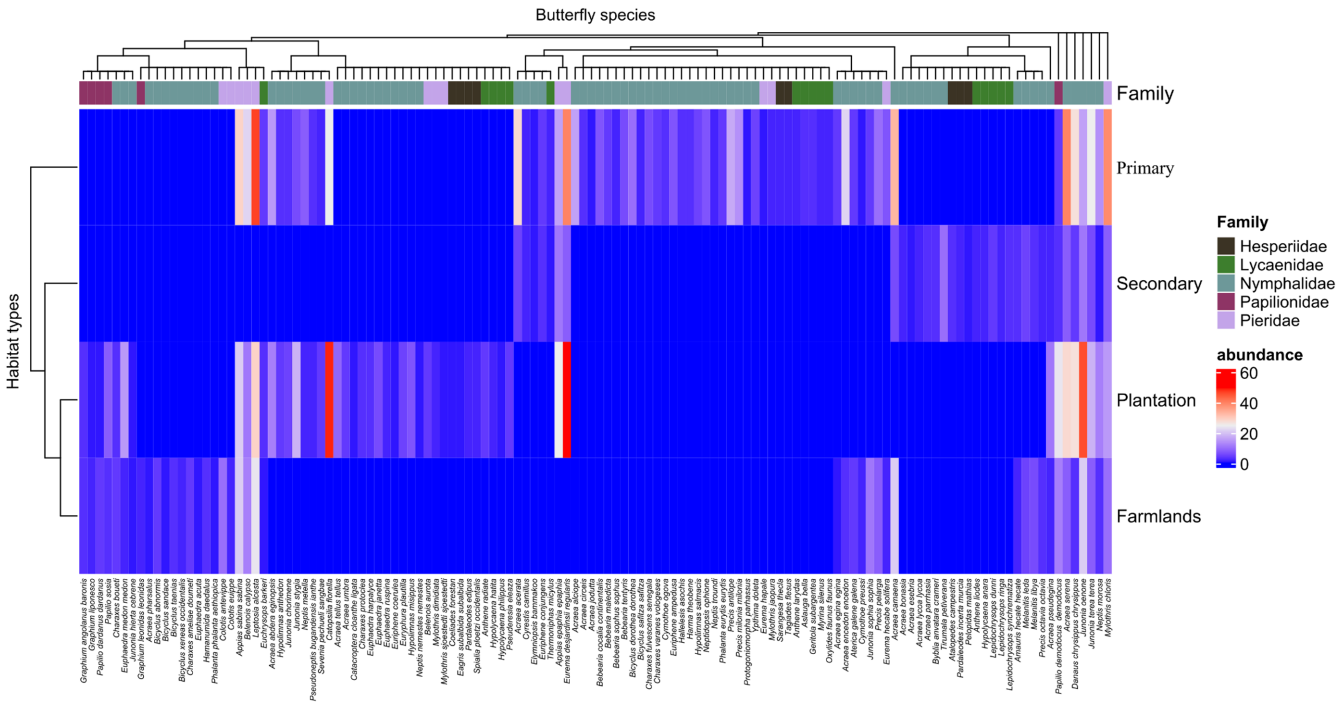

FIGURE 1 Community variation of butterflies across habitat types. The butterfly species composition in farmlands showed high similarity with plantation and secondary forests, but low similarity with Primary forests. Plantation forests had low similarity with both Secondary and Primary forests, while the Primary forests had moderate similarity with the Secondary forests.

4 | DISCUSSION

A wide range of habitats within the studied area harbour diverse butterfly communities, which may be attributed to many butterfly species being attracted to different habitats, including those that human activities have modified to support flower-producing plants. Despite variations in habitat types, the overall evenness of butterfly species distribution was consistently high, with a value of over 88% across all habitats surveyed. Nevertheless, forested habitats supported more butterfly species than disturbed landscapes. Consequently, conservation efforts to preserve butterfly diversity should prioritise habitats with a greater potential for supporting a variety of species (Berg et al., 2011).

The status of many habitats is largely influenced by human disturbance due to their use such that habitats with less human disturbance tend to have a higher butterfly species assemblage than those with higher disturbances. In Nigeria's tropical forest vegetation zone, few Primary forests exist that are less disturbed, as most are being exploited for agriculture to provide food for the growing human population. As a result, some of the surveyed habitats support fewer biological resources. The high species richness and abundance recorded in the Primary forests may be attributed to their high habitat heterogeneity and less human disturbance. Two butterfly families, Nymphalidae and Pieridae, appear to be more adapted to Primary forest habitats than other butterfly families recorded (Ojianwuna & Enwemiwe, 2021). These families are known to be abundant and co-exist in various locations and habitats (Paul & Sultana, 2021; Velasquez Blanco, 2022).

Furthermore, many of the species of the nymphalid and pierid families are less active and high flying than the papilionid family. Hence, they may tend to persist within habitats with dense vegetation. The availability of open space for the active and high-flying butterfly species and the abundance of flowering plants in open habitats (plantation forests and farmlands) may have supported the abundance of the Family Papilionidae. No butterfly species of the family Hesperiidae being recorded in farmlands could be a play-out of a healthy habitat preference behaviour among species of that family (Summerville & Clampitt, 1999). Considering that species of the hesperiid family were recorded in the other three habitats, except farmland, hesperiid species may need to be better adapted to highly disturbed habitats (Garcia-Barros & Fartmann, 2009; Porter, 1992). Continuous cultivation alters the vegetation makeup of farmlands,

reduces the number of flowering plants available to the butterfly species, and could make such habitat patches unattractive to the hesperiid species. However, there is evidence of a hesperiid (*Pyrgus ruralis lagunae* adults) thriving in open habitats and forest edges but not within a grazing enclosure (Marschalek et al., 2017).

The butterfly species composition in farmland being most similar to those in plantation forests, followed by the secondary forests, and least similar to the Primary forests, suggest that farmlands, which is often highly disturbed, may still support some butterfly species that are adapted to plantation forest habitats, while Primary forests harbour a unique set of butterfly species that are not found in other habitats. These findings highlight the importance of preserving Primary forests to maintain butterfly diversity in the region. The butterfly species supported by farmlands and plantation habitats may be more adapted to disturbed areas (Fotang, Bröring, Roos, Enoguanbhor, Abwe, et al., 2021; Fotang, Bröring, Roos, Enoguanbhor, Dutton, et al., 2021).

Considering the habitat sizes and their butterfly species assemblages across the studied area, it could be said that butterfly assemblages in habitats are more influenced by anthropogenic disturbances and least by their sizes (van Schalkwyk et al., 2021). This is because Secondary forests, which cover a greater percentage of the landmass across South-East Nigeria, have the least diversity and abundance of butterflies compared to other habitats under study. Furthermore, the variations in butterfly assemblages between habitat pairs also show that butterfly species supported by various habitats vary greatly, and this could be dependent on the diversity of resources provided by the habitats, as well as the level of human disturbance occurring in these habitats (Biswas & Mallik, 2010). Many species may have adapted to certain habitats with a high level of human disturbance due to their natural attraction to certain plant communities, irrespective of the habitat condition (Sher & Hyatt, 1999).

In conclusion, habitat disturbance significantly impacts butterfly species assemblage in the Afrotropical region, which could possibly lead to declines in population abundance, changes in community composition and structure, alterations in environmental conditions, and increased vulnerability to extinction. Therefore, conservation efforts that prioritise protecting and restoring native habitats are critical to maintaining the diversity and health of butterfly communities in the region.

AUTHOR CONTRIBUTIONS

LNO conceived and designed the study as well as collected the data; ECN analysed the data; LNO and ECN wrote the manuscript; AUE supervised the entire work and proof-read the manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declare that they have no conflict of interest.

DATA AVAILABILITY STATEMENT

Data supporting this study are available upon request from the correspondent author.

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REFERENCES

Attiwilli, S., Karmakar, T., Isvaran, K., & Kunte, K. (2022). Habitat preference and functional traits influence responses of tropical butterflies to varied habitat disturbance. *International Journal of Tropical Insect Science*, *42*, 1–10.

Berg, Å., Ahrné, K., Öckinger, E., Svensson, R., & Söderström, B. (2011). Butterfly distribution and abundance is affected by variation in the Swedish forest-farmland landscape. *Biological Conservation*, *144*(12), 2819–2831.

Biswas, S. R., & Mallik, A. U. (2010). Disturbance effects on species diversity and functional diversity in riparian and upland plant communities. *Ecology*, *91*(1), 28–35.

Brown, K. S., Jr., & Hutchings, R. W. (1997). Disturbance, fragmentation, and the dynamics of diversity in Amazonian forest butterflies. In *Tropical forest remnants: Ecology, management, and conservation of fragmented communities* (p. 632). University of Chicago Press.

Fotang, C., Bröring, U., Roos, C., Enoguanbhor, E. C., Abwe, E. E., Dutton, P., & Birkhofer, K. (2021). Human activity and Forest degradation threaten populations of the Nigeria–Cameroon chimpanzee (*pan troglodytes ellioti*) in Western Cameroon. *International Journal of Primatology*, *42*(1), 105–129.

Fotang, C., Bröring, U., Roos, C., Enoguanbhor, E. C., Dutton, P., Tédonzong, L. R., & Birkhofer, K. (2021). Environmental and anthropogenic effects on the nesting patterns of Nigeria-Cameroon chimpanzees in North-West Cameroon. *American Journal of Primatology*, *83*(9), e23312.

Garcia-Barros, E., & Fartmann, T. (2009). Butterfly oviposition: Sites, behaviour and modes. In J. Settele, M. Konvicka, & T. Shreeve (Eds.), *Ecology of butterflies in Europe* (pp. 29–42). *Cambridge University Press*.

Gu, Z. (2022). Complex heatmap visualization. *iMeta*, *1*(3), e43.

Habel, J. C., Seibold, S., Ulrich, W., & Schmitt, T. (2018). Seasonality overrides differences in butterfly species composition between natural and anthropogenic forest habitats. *Animal Conservation*, *21*(5), 405–413.

Hamer, K. C., Hill, J. K., Lace, L. A., & Langan, A. M. (1997). Ecological and biogeographical effects of forest disturbance on tropical butterflies of Sumba, Indonesia. *Journal of Biogeography*, *24*(1), 67–75.

Hammond, P. C., Lepczyk, C. A., Flaspohler, D. J., & Langham, G. M. (2019). Assessing habitat suitability for monarch butterflies in the Midwest US using an agent-based model. *Ecological Applications*, *29*(1), e01803.

Kocher, S. D., & Williams, E. H. (2000). The diversity and abundance of North American butterflies vary with habitat disturbance and geography. *Journal of Biogeography*, *27*, 785–794.

Larsen, T. B. (2005). *Butterflies of West Africa* (Text Volume and Plate Volume). Apollo Books.

Maechler, M., Rousseeuw, P., Struyf, A., Hubert, M., & Hornik, K. (2022). *Cluster: Cluster analysis basics and extensions*. [software]. https://CRAN.R-project.org/package=cluster

Marschalek, D. A., Faulkner, D. K., & Deutschman, D. H. (2017). Livestock grazing shapes the vegetation structure and subsequent habitat use by the endangered skipper *Pyrgus ruralis lagunae* (Lepidoptera: Hesperiidae). *Environmental Entomology*, *46*(3), 445–453.

Ojianwuna, C. C., & Enwemiwe, V. N. (2021). Spatial distribution of butterflies in different macrohabitat in a university campus in Southern-Nigeria. *International Journal of Tropical Insect Science*, *41*, 1–12.

Oksanen, J., Simpson, G., Blanchet, F., Kindt, R., Legendre, P., Minchin, P., O'Hara, R., Solymos, P., Stevens, M., Szoecs, E., Wagner, H., Barbour, M., Bedward, M., Bolker, B., Borcard, D., Carvalho, G., Chirico, M., de Caceres, M., Durand, S., … Weedon, J. (2022). *vegan: Community ecology package*. R Package version 2.6–4, https://CRAN.R-project.org/package=vegan

Paul, M., & Sultana, A. (2021). Diurnal activity budget of some notable butterfly (Insecta: Lepidoptera) species of selected Familiein Delhi, India: A comparative study. *Indian Forester*, *147*(10), 1000–1005.

Pollard, E. (1977). A method for assessing the abundance of butterflies. *Biological Conservation*, *12*, 115–134.

Porter, K. (1992). Eggs and egg-laying. In R. L. H. Dennis (Ed.), *The ecology of butterflies in Britain* (pp. 46–72). Oxford University Press.

R Core Team. (2021). *R: A Language and Environment for Statistical Computing*. R foundation for statistical computing.

Satterfield, D. A., Maerz, J. C., & Altizer, S. (2015). Loss of migratory behaviour increases infection risk for a butterfly host. *Proceedings of the Royal Society B: Biological Sciences*, *282*(1803), 20141734. https://doi.org/10.1098/rspb.2014.1734

Sher, A. A., & Hyatt, L. A. (1999). The disturbed resource-flux invasion matrix: A new framework for patterns of plant invasion. *Biological Invasions*, *1*(2), 107–114.

Spitzer, K., Jaroš, J., Havelka, J., & Lepš, J. (1997). Effect of small-scale disturbance on butterfly communities of an Indochinese montane rainforest. *Biological Conservation*, *80*(1), 9–15.

Spitzer, K., Novotny, V., Tonner, M., & Leps, J. (1993). Habitat preferences, distribution and seasonality of the butterflies (Lepidoptera, Papilionoidea) in a montane tropical rain forest, Vietnam. *Journal of Biogeography*, *20*, 109–121.

Summerville, K. S., & Clampitt, C. A. (1999). Habitat characterization of five rare insects in Michigan (Lepidoptera: Hesperiidae, Riodinidae, Satyridae; Homoptera: Cercopidae). *The Great Lakes Entomologist*, *32*(3), 13.

Thomas, J. A., Telfer, M. G., Roy, D. B., Preston, C. D., Greenwood, J. J. D., Asher, J., Fox, R., Clarke, R. T., & Lawton, J. H. (2004). Comparative losses of British butterflies, birds, and plants and the global extinction crisis. *Science*, *303*(5665), 1879–1881.

van Schalkwyk, J., Gaigher, R., Pryke, J. S., & Samways, M. J. (2021). Within-corridor heterogeneity is more important than corridor design for maintaining butterfly functional and taxonomic diversity. *Journal of Applied Ecology*, *58*(12), 2734–2746.

Velasquez Blanco, D. (2022). *Efectos de la urbanización en la comunidad de mariposas diurnas y escarabajos coprófagos de los ecoparques de Manizales*, Colombia.

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